

KEYWAVE

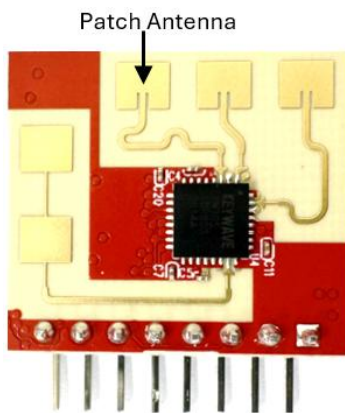
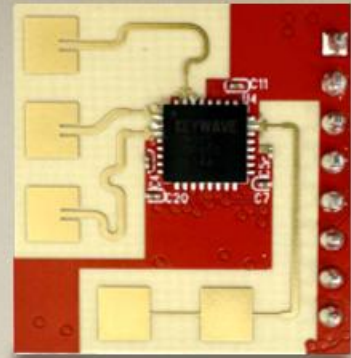
Keywave-KW307

Human Presence Sensor Module

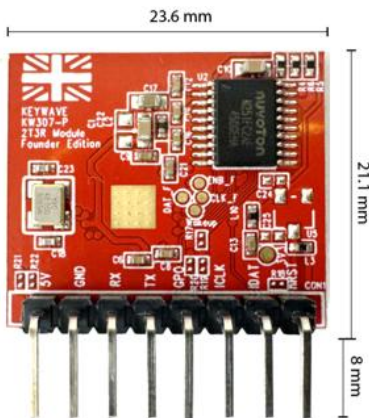
- Revolutionary new Radar technology
- Ultra-low power consumption
- Detects multiple and stationary People
- Reduces false alarms

User Guide and Datasheet

Version 2.2 | 2026-07-29, Keywave Technology Ltd.



Front view
(antenna facing forward)



Back view



Side View

Product Overview

The **KW307** is a compact 24 GHz RF-based **human presence sensor module** designed to detect both moving and stationary people in indoor and outdoor, embedded sensing applications. With integrated antenna, UART target reporting, configurable detection range, adjustable field of view, and support for multi-target tracking of up to three people, the KW307 enables reliable presence-based control for smart buildings, automation, energy-saving systems, and human-machine sensing applications.

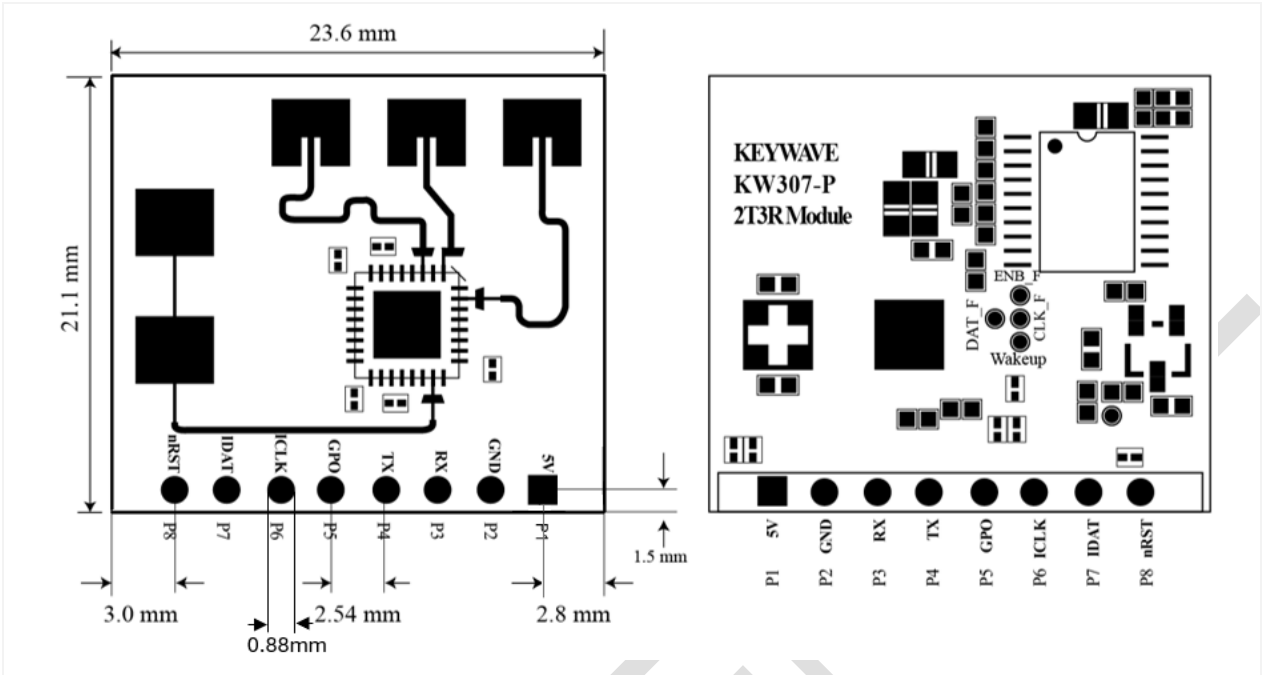
Key Features

- 24 GHz RF-based human presence sensing module
- Detects both moving and stationary humans
- Multi-target tracking of up to four people
- Reports target distance and angle information via UART
- Configurable detection range, field of view, gain, sensitivity, and hold time
- Typical 100° field of view, configurable up to 120°
- Distance accuracy up to ± 5 cm and angle accuracy up to $\pm 3^\circ$
- Strong immunity to fan clutter, wind, and vibration
- Supports 3.6V ~ 5V power input
- PC GUI tool for configuration, scenario presets, and real-time detection monitoring

Typical Applications

- Smart lighting and HVAC control, Smart building automation
- Room and desk occupancy detection
- Energy-saving monitoring systems, Smart home and appliance control
- Washroom automation, including urinal and hand dryer sensing
- Human-machine interface and embedded presence detection
- Evaluation and prototyping for RF sensing products
- Recommended clearance zone: at least 10mm in total (front and back).

Module Dimensions & Pin Definitions



Description	Min.	Typ.	Max.	Unit
Module Length	23.58	23.6	23.62	mm
Module Width	21.08	21.1	21.12	mm
Module Thickness	1.57	1.6	1.63	mm
Hole Dimension	0.875	0.88	0.9	mm
Pin Header Diameter		0.6		mm

Pin	Symbol	Type	Description
P1	5V	Power	5V power supply
P2	GND	Ground	Ground
P3	RX	Input	UART Rx signal
P4	TX	Output	UART Tx signal
P5	GPO	Output	General-purpose output. *it can be the human flag output.
P6	ICLK	Input	Programming clock signal. *It can be left unconnected in normal application.
P7	IDAT	Input	Programming data signal. *it can be left unconnected in normal application.
P8	nRST	Input	System reset signal (active low) *it can be left unconnected in normal application.

- 5V Power Mode: Supply voltage is regulated by an on-chip LDO to power the internal core.

Block Diagram

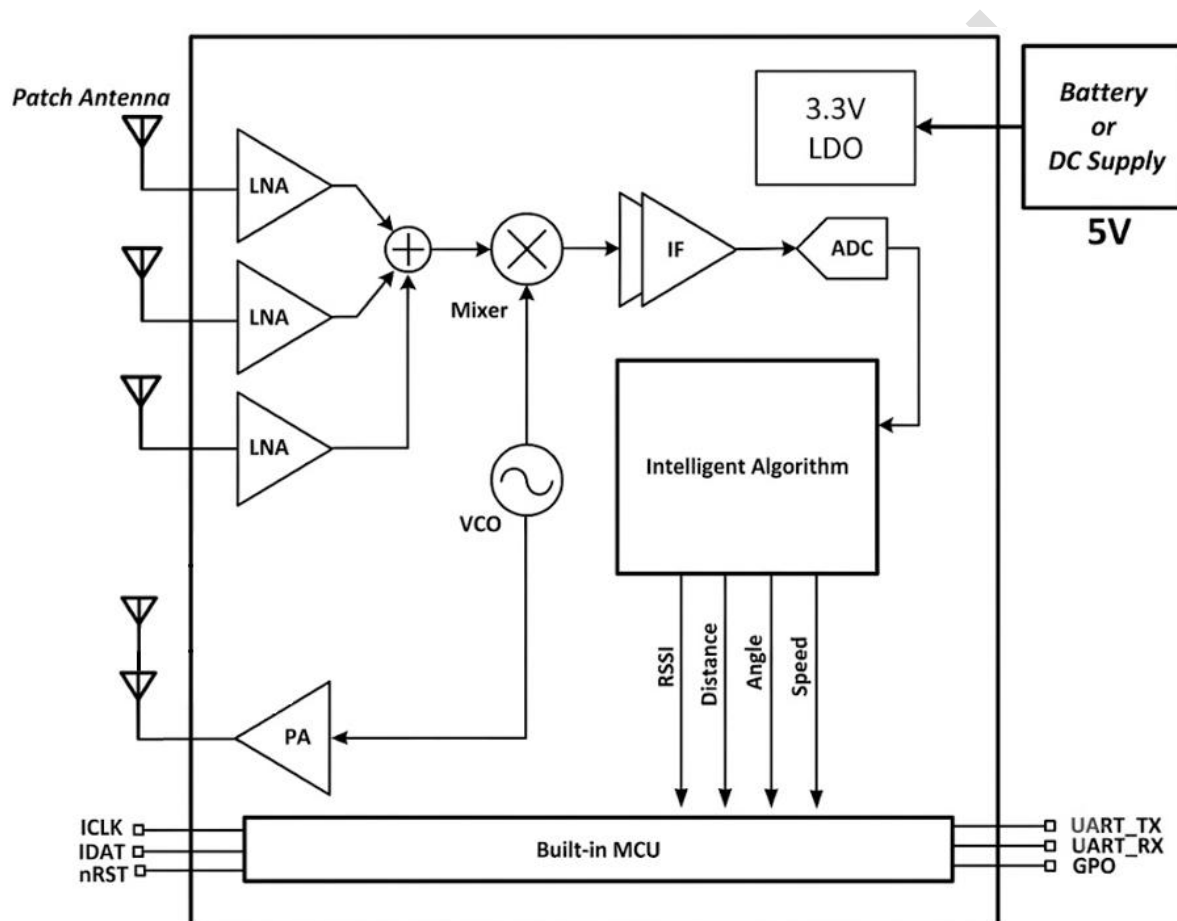


Figure : KW307 Block Diagram

Electrical Specification

Absolute Maximum Ratings

Characteristic	Symbol	Min.	Typ.	Max.	Units
5V supply voltage	V_{DD5V}	-0.3	5.0	6.0	V
Digital I/O voltage	V_{DIO}	-0.3	3.3	3.6	V
Digital I/O maximum current capability	$I_{DIO,max}$			10	mA
Ambient Temperature	T_A	-25		85	°C

* GPO cannot directly drive a high-current load; it requires a power MOS switch to do so.

DC Characteristics

Characteristic	Symbol	Min.	Typ.	Max.	Units
Supply voltage via 5V pin	V_{DD5V}	3.6	5.0	6.0	V
Operation Current @ 20Hz update rate	I_{DC}		4.0		mA
Operation Current @ 0.5Hz update rate	I_{DC}		0.8		mA

RF Characteristics

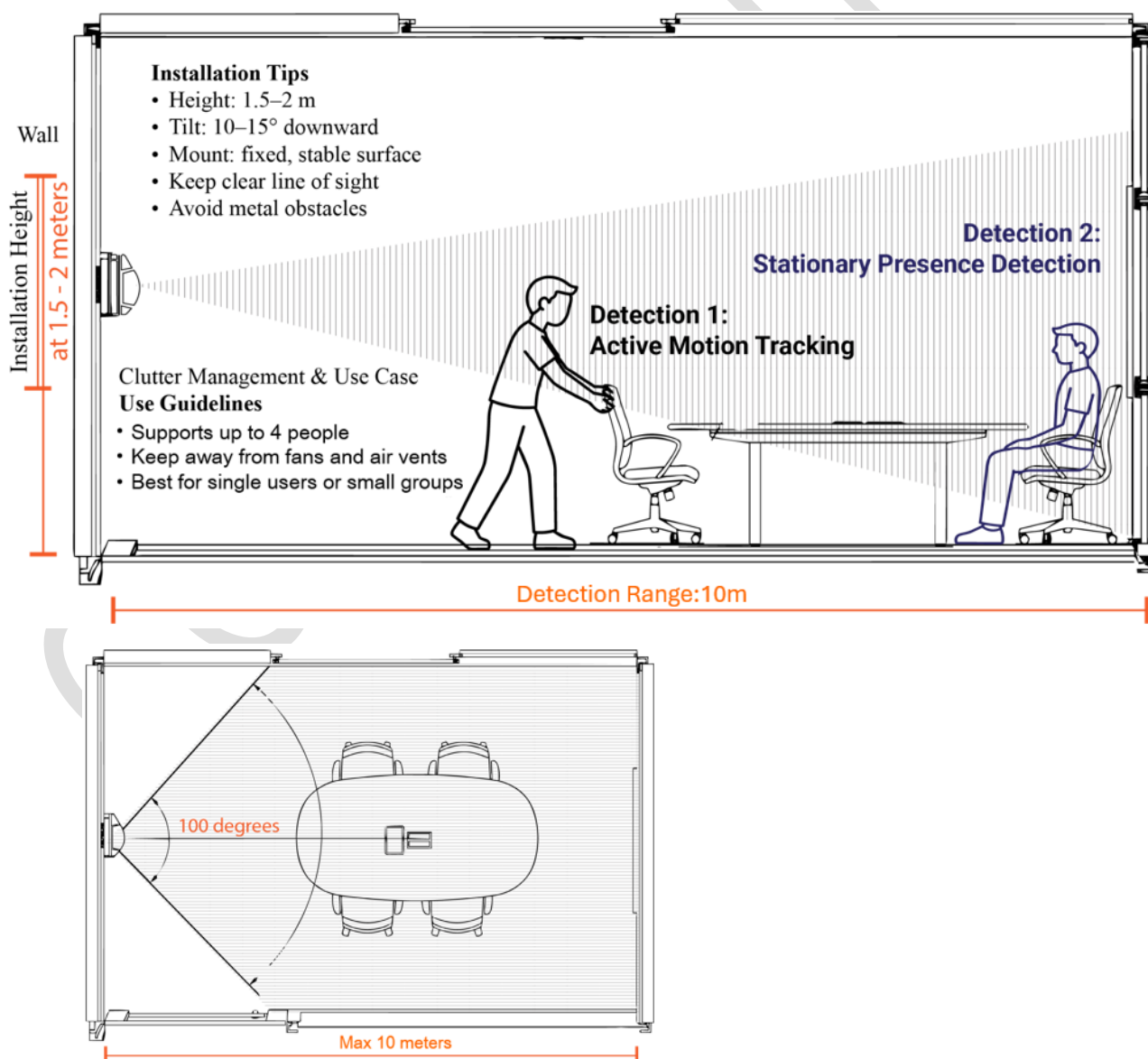
Characteristic	Symbol	Min.	Typ.	Max.	Units
Operation frequency	F_{RF}	24.05		24.25	GHz
RF output power	P_{RF}		-2		dBm
Detection range	Range		10	20*	m
Field of View	FOV		100		degrees

*Horizontal alignment at 1m height.

*For detection requirements up to 20 meters, please contact Keywave representative

Mounting Suggestion

- **Optimal Installation Height:** For best results, mount the sensor at a height of 1.5m to 2m above the floor.
- **Avoid Obstructions:** Ensure a clear line of sight. Avoid placing the sensor behind metallic objects or thick glass, as these may block or attenuate the radar signal.
- **Directional Alignment:** If the target area is close to the mounting point tilt the sensor slightly downward (approx. 10° – 15°) to maximize FOV coverage.
- **Stability:** Mount the sensor on a rigid surface. Although the KW307 has superior immunity to vibration, a stable mount ensures the highest possible accuracy for stationary detection.
- **Clutter Management:** Position the sensor away from large moving mechanical parts (such as large industrial fans) to maintain the best tracking performance for multiple people.

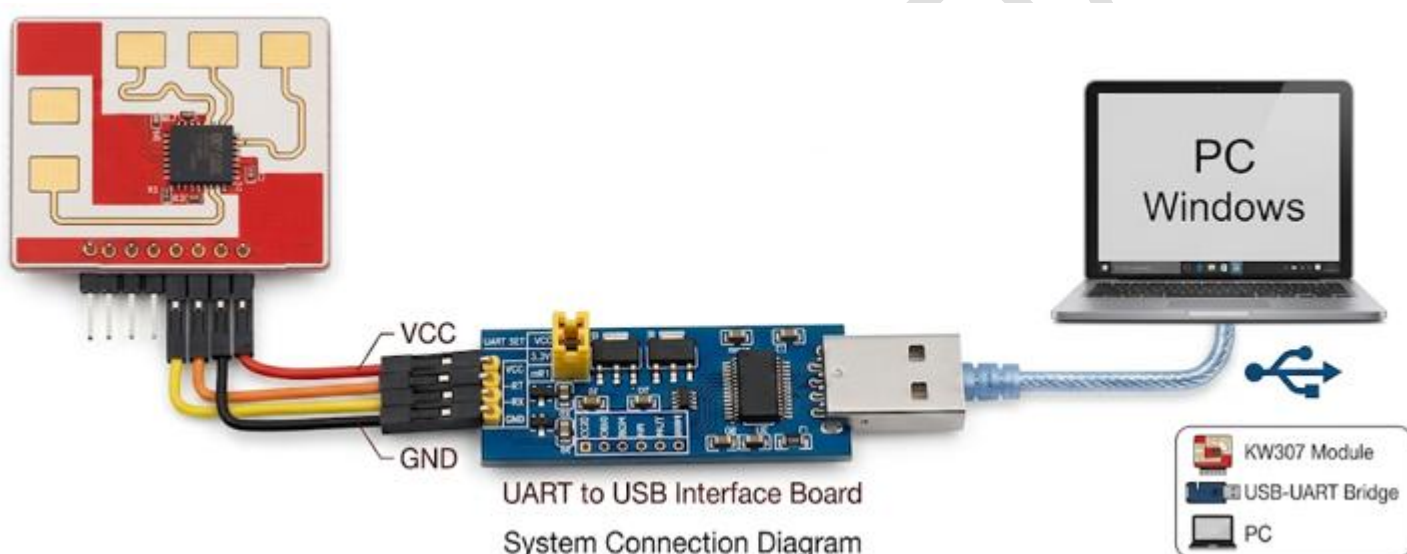


Configuration and monitoring tool

The KW307 configuration and monitoring tool is a GUI application designed to help you get up and running as fast as possible.

The tool allows you to:

- Configure sensor detection parameters, making it easy to tune the KW307 to your application
- Choose pre-set tuning parameters for common applications.
- Monitor real-time detection status, on an intuitive graphical display.



- The configuration tool uses a USB to UART adapter to communicate from a Windows PC to the KW307.
- Note the Pin spacing for the KW307 is 2.54mm.
- UART RX/TX interface must be 3.3V. (Not 5V Tolerant).

CAUTION :

- When connecting the adaptor, or bridge board, ensure you use a data-capable USB cable, not a charge only cable.
- Take care when connecting the cable as connection errors may permanently damage the hardware module.
- The configuration tool software can be found at: <https://keywavetech.co.uk/products>

Step1 : Select "COM Port"

Step2 : Click "Connect"
(to connect com port)

Step3 : Click "Check FW"
(It should be v2.6.1)

Step4 : Select "Application"

Step5 : Click "Enter"
(Setup the setting)

UART / COM Control Center

Port COM11 Baud 115200 Connect Disconnect Connected

KEYWAVE KW307

System

Check FW Factory Reset

Initial Save Setting

Customer Control Panel

Application Office

Gain 5: +12 dB

Room range (cm) 50 1000

Field of view/Shrink Step 120 8

Motion sensitivity 2 ☐ byp_spd

Stationary sensitivity 1

Flag Hold time (s) 11

Mag floor mov / stat 350 100

Moving trk max / show 32 8

Stationary trk max / show 64 8

☒ Force statio.(cm/deg) 150 40

☒ Max still time (s) 20

☒ Adv clutter rej(sth/dth) 300 20

☐ Power down (sleep)

Enter (Apply Settings)

KEYWAVE KW307 Radar

2m 4m 6m 8m 10m

-90deg -75deg -60deg -45deg -30deg -15deg 0deg 15deg 30deg 45deg 60deg 75deg 90deg

Human Flag

1

Output (presence + targets)

	Dist(cm)	Ang(deg)	Mag	State
Moving 0	82	-21.8	2030	Shown
Moving 1	279	1.5	0	Check
Stationary 0	256	-29.6	5748	Hidden
Frame #	8723	Radar fade threshold	16	

☒ Same Color ☐ en_Fade ☐ show_DistAng ☐ show_mag ☒ Plot_History ☐ Flip_Horizontally RadarPlotMaxDist 1000 cm Clear

Ready: Pick a COM port and Connect.
Opened COM11 @ 115200
Connected.
MCU firmware: v2.6.1
Application preset: Office (press Apply to send)
gain 5 (rc=0)
range 50~1000 cm (rc=0)
fov 120 / shrink 8 (rc=0)
motion sens 2 byp 0 (rc=0)
stationary sens 1 (rc=0)
flag hold 11 s (rc=0)
mag floor 350/100 (rc=0)
moving trk 32/8 (rc=0)
still trk 64/8 (rc=0)
force stat 1 (150 cm / 40 deg) (rc=0)
max still 1 (20 s) (rc=0)
clutter 1 (50 / 20) (rc=0)
update rate 0 (cancel) (rc=0)
power down 0 (rc=0)
Applied.

Module Active

- The pre-loaded application scenarios come with pre-defined optimal parameters, while still allowing users to fine-tune settings for specific applications. Please click "Enter" after you complete the desire settings.

Function	Range	Description
Application Mode	5 scenarios	Outdoor, Office, Desk, Urinal, Hand_dryer
Gain	-3db ~ +21db	Adjust radar signal strength. (+21dB: the maximum adjustable signal strength.)
Room Dist (min/max)	10 ~ 1200	Set the range size for radar detection. (cm)
Field of view (FOV)	0 ~ 120	Set the radar field of view. (degree)
FOV Shrink step	0 ~ 25	The FOV narrows degree per 100cm. default:8 means shrink 8deg/100cm
Motion Sensitivity	L0 ~ L9	The ability to detect a moving human. L0: Most sensitive (i.e., Wave hands or shake heads) L9: Least sensitive (i.e., Walk one step)
Stationary Sensitivity	L0 ~ L9	The ability to detect subtle movement of a stationary human. L0: Most sensitive (i.e., Breath) L9: Least sensitive (i.e., Wave hands or shake heads)
Flag Hold time	0 ~ 3000	Set the hold time of Human Flag as human leaves detection area (Sec).

Function	Range	Description
Mag floor (threshold) motion	0 ~ 2000	Signal strength above the mag threshold will be identified as valid targets for a moving human. (Environment-dependent, default: 350)
Mag floor (threshold) stationary	0 ~ 2000	Signal strength above the mag threshold will be identified as valid targets for a stationary human. (Environment-dependent, default: 100)
Motion tracker max / show	1 ~ 64	Max: The maximum number of frames a moving target survives in the tracker plot without receiving any new detection updates. Show: The detection count threshold required for a newly initialized target to transition into the "Shown" (active/confirmed) state.
Stationary tracker max / show	1 ~ 64	Max: The maximum number of frames a stationary target survives in the tracker plot without receiving any new detection updates. Show: The detection count threshold required for a newly initialized target to transition into the "Shown" (active/confirmed) state.
Force stationary (cm/deg)	10~1000 cm/ 0~120 deg	Setup a forward window to keep the presence signal suitable while a still target stands, so brief detection dropouts don't flicker presence while a person sits very still. default: 150cm/40deg * It needs to tick the enable box to turn on this function.
Max still time	0 ~ 3000	The tracking dot on the radar plot will vanish after the target remains stationary for the configured seconds . (Sec) * It needs to tick the enable box to turn on this function.
Adv clutter reject (sth/dth)	0~1000 sth/ 0~50 dth	Reject repetitive background motion (AC, machinery, ...) so it is not reported as presence. Turn it off to report every moving target. default: 300sth / 20dth * It needs to tick the enable box to turn on this function.
Power down (sleep)	-	Power down the sensing function to save power. * It needs to tick the enable box to turn on this function.

- Configuration parameters can be saved to persistent memory; they will remain intact after a power cycle and be loaded at the next startup.

Function	Description
Factory Reset	Clicking "Factory Reset" will clear the saved NVM (Non-Volatile Memory) settings in the MCU and restore the system to its factory firmware defaults.
Save Setting	Clicking "Save Setting" will overwrite the MCU's stored NVM configuration with the current settings. The saved configuration remains effective even after a power outage.
Initial	Loads the configuration parameters currently stored in the MCU's NVM.

Contact: Q@keywavetech.co.uk

Item	Description
Product	KW307 Human Presence Sensor Module
Part number	KW307
Module variant	KW307-P
Product name	KW307 Human Presence Sensor Module
Sensor type	24 GHz RF-based human presence sensor module
Interface	UART, GPO, programming interface pins
Supply options	3.6V ~ 5V supply input
Package / module type	PCB module with integrated antenna
Module dimensions	Approximately 23.6 mm × 21.1 mm
Pin header	2.54 mm pitch on KW307 module
Hardwar version	KW307-P-M1
Firmware version	v2.6.3
GUI revision	vB.1
Datasheet version	v2.2
Evaluation support	PC GUI configuration and monitoring tool

For the latest updates, visit our website: www.keywavetech.co.uk

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